



NATIONAL OCCUPATIONAL STANDARD FOR SOLAR PV INSTALLER

APPROVING AUTHORITY

This National Occupational Standard has been prepared and published under the authority of the Zambia Qualifications Authority Board on 21st September 2021.

ZAMBIA QUALIFICATIONS AUTHORITY

The Zambia Qualifications Authority Act No. 13 of 2011 was enacted by the Government of the Republic of Zambia to ***“provide for the development and implementation of a national qualifications framework; establish the Zambia Qualifications Authority; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing”***. Among other functions, ZAQA is responsible for ***determining national standards for any occupation***, through various sector specific National Occupational Standards Development Teams (NOSDTs).

REVISION OF NATIONAL OCCUPATIONAL STANDARDS

National Occupational Standards shall be revised every after **5 years**, or whenever necessary, by the issue of either amendments or of revised editions. It is important that users of National Occupational Standards (NOS) should ascertain that they are in possession of the latest amendments or editions.

NOS DEVELOPMENT TEAM RESPONSIBLE

This National Occupational Standard was prepared by the Energy National Occupational Standards Development Team, upon which the following organisations were represented:

1. Copperbelt University;
2. Department of Energy - Ministry of Energy;
3. Energy Regulation Board;
4. Kafue Gorge Regional Training Centre;
5. Mines Safety Department - Ministry of Mines and Minerals Development;
6. Ministry of Labour and Social Security Services;
7. Muhanya Solar;
8. Northern Technical College;
9. Rural Electrification Authority;
10. SNV Zambia;
11. University of Zambia;
12. Zambia Qualifications Authority.

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FOREWORD

The Zambia Qualifications Authority (ZAQA) is a statutory body under the Ministry of Higher Education established by ZAQA Act No. 13 of 2011 to ***“provide for the development and implementation of a national qualifications framework; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing”***.

Among other functions, ZAQA is responsible for ***“determining national standards for any occupation”***, through various sector specific National Occupational Standards Development Teams (NOSDTs) of experts composed of representation from appropriate authorities, government departments, industry, academia, regulators, consumer associations and non-governmental organizations, etc.

This National Occupational Standard (NOS) has been developed by the Energy National Occupational Standards Development Team in accordance with the procedures and guidelines of ZAQA. All users should ensure that they have the latest edition of this publication as National Occupational Standards are revised from time to time.

This NOS shall be used by, among others, industry, employers, quality assurance bodies, awarding and professional bodies and education and training institutions, as a benchmark to identify training needs, develop job profiles/descriptions, develop curricula and learning programmes, in various sectors where the occupation exists. In the Energy sector, demonstration of competence against this NOS may be required in order to run a business or practice a craft or profession.

JUSTIFICATION

Solar photovoltaic (PV) installers are key to the process of solar PV equipment installation, operations and maintenance. They are specialized for civil, mechanical and electrical integration of Solar Photovoltaic equipment. Furthermore, the PV industry is rapidly maturing because of worldwide environmental concerns and its energy production potential from the sun.

With the foregoing in mind, various establishments will require solar PV installers to design, install, operate and maintain solar PV systems. Solar PV installers use specialized skills to design, install, operate and maintain residential and commercial solar projects. They are responsible for safely installation of solar PV equipment and insuring that the system works.

This National Occupational Standard highlights core knowledge, skills, competences and personal attributes that Solar PV Installers must possess to be successful in their jobs.

ACRONYMS AND ABBREVIATIONS

A-h.	Ampere-hour
CS	Core Skill
DoD	Depth of Discharge
I-V	(I) Current – (V) Voltage
MLPE	Module Level Power Electronics
MPPT	Maximum Power Point Tracker
NOS	National Occupational Standard
NOSDT	National Occupational Standards Development Team
OK	Organisational Knowledge
OCPD	Over-Current Protection Device
OHS	Occupational Health and Safety
PC	Performance Criteria
PdM	Predictive Maintenance
PS	Professional Skill
RCM	Reliability Centred Maintenance
RK	Regulatory Knowledge
RPL	Recognition of Prior Learning
SCADA	Supervisory Control and Data Acquisition
SLD	Single Line Diagrams
SPVI	Solar Photovoltaic Installer
TK	Technical Knowledge

ZAQA	Zambia Qualifications Authority
ZQF	Zambia Qualifications Framework

GLOSSARY OF TERMS

For the purposes of this NOS, the following terms and definitions shall apply:

Core Skills/Generic Skills: are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.

Function: is an activity necessary for achieving the key purpose of the sector, occupation, or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of NOS.

Job Title: defines a unique set of functions that together form a unique employment opportunity in an organisation.

Knowledge and Understanding: are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

National Occupational Standards (NOS): are statements of the standards of performance individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding. They are precise descriptions of what an individual is expected to be able to do in his/her work role.

National Occupational Standards (NOS) Code: is a unique reference code that identifies a NOS.

National Occupational Standards Development Team (NOSDT): means an established group of national stakeholders/experts responsible for the development of National Occupational Standards within a specific economic sector or occupation.

Occupation: is a set of job roles, which perform similar/related set of functions in an industry.

Organisational Context: includes the way the organisation is structured and how it operates, including the extent of operative knowledge that managers have in their relevant areas of responsibility.

Performance Criteria: are statements that together specify the standard of performance required when carrying out a task.

Scope: is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.

Sector: is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.

Sub Sector: is derived from a further breakdown based on the characteristics and interests of its components.

Technical Knowledge: is the specific knowledge needed to accomplish specific designated responsibilities.

Unit Title: gives a clear overall statement about what the incumbent should be able to do.

1. OVERVIEW

This is an introductory section providing a brief summary and specific information or commentary about the content of the NOS and the targeted sector and occupation to help the user judge whether it is relevant to them.

NOS Code	NOS.SPVI.01
Occupation	Installation, Operation and Maintenance
Job Title	Solar PV Installer
Job Description	Solar PV Installers are responsible for checking, adapting, implementing, configuring, electrical trouble shooting, installing, inspecting, testing, and commissioning different components of photovoltaic systems, that meet the performance and reliability needs of customers by incorporating quality craftsmanship and complying with all applicable codes, standards, and safety requirements.
Job Purpose	System design, installation, operation and maintenance of solar PV system equipment.
ZQF Level	Level 6
Sector	Energy
Sub sector	Renewable Energy
Other Economic Sector(s) in which the Occupation is Practiced	Academia, Research and Development, Regulatory and Standards Agencies, Agriculture, Construction, Mining Engineering, Manufacturing, Health, Telecommunication, Oil and Gas, Rural Electrification Authority.
Other Similar Jobs that can be performed by the Solar PV Installer	Data logging, disposal of expired PV equipment.
Minimum Educational Job Entry Qualification(s)	Diploma
Practicing License Requirements (if any)	Engineering Registration Board Practicing Licence
Training/RPL	- Occupational Health and Safety Training - Workshops in solar PV technology
Minimum Job Entry Age	Anyone qualified to be employed under the laws of the country (21 years)
Prior Experience	At least 3 months internship in the Solar industry.
Performance Criteria	As described in the Units under Section 4.

2. SCOPE

This National Occupational Standard specifies the fundamental knowledge and understanding, skills and competences that a Solar PV installer must possess to be successful in his/her job role. It is applicable to Solar PV Installers working in public or private organisations or self-employed within or outside the Solar energy sector.

3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)

This job requires an individual to possess:

- Creativity
- Problem solving skills
- Analytical skills
- Mathematical skills
- Integrity
- Interpersonal skills
- Commercial Awareness
- Attention to detail

4. UNITS AND ELEMENTS

This National Occupational Standard is divided into five (05) Units representing the tasks that a job holder should undertake in his/her day to day work. Each unit is further broken down into elements depicting the number of activities to be carried out for the successful execution of a particular task.

UNIT 1 [This unit covers the skills, knowledge and understanding required by a solar PV installer to be proficient in carrying out system design of a Solar PV System by means of site surveying and load analysis.]

Unit No.	01
Unit Title	Planning and Designing of a Solar PV System
Description	This unit is about planning and designing of Solar PV system.
Scope	This unit covers the following: • Load analysis • Site survey • System design
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Identify load to be connected to Solar PV System	To be competent, the individual must be able to: PC1. Assess the load to be run on Solar PV system. PC2. Prepare a load profile. PC3. Document the site survey variables and complete the checklist/site survey form.
Assess the site conditions	To be competent, the individual must be able to: PC4. Understand the location of installations and optimise route plan. PC5. Assess the site level pre-requisites for solar panel installations. PC6. Check for any shading obstacles. PC7. Decide on the type of mounting to be constructed. PC8. Inform the customer for any civil construction to be undertaken for installing the solar modules. PC9. Prepare a site map of the location where installation has to be carried out.
Size the various components of a solar system	To be competent, the individual must be able to: PC10. Determine the size of solar PV array. PC11. Determine the size of the battery bank. PC12. Determine the size of inverter/charge controller. PC13. Determine the size of the cables.
Knowledge and Understanding (K)	
A. Organisation-al Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company's Installation Policy. OK2. Company's Customer Support Policy. OK3. Company's documentation policy. OK4. Documenting information using appropriate corporate forms. OK5. Company's reporting structure. OK6. Organization culture.

B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of : TK1. Solar PV technology. TK2. Types of solar modules. TK3. Components of a solar PV system. TK4. Definition of the technical terms: energy and power, solar cell, solar module, module string, solar array. TK5. Relevant concepts of coordinate geometry. TK6. Irradiation and irradiance and their units and symbols. TK7. Series and parallel connections of solar modules, tilt angle, geographical north, magnetic north, azimuth angle, orientation and shading. TK8. Solar radiation maps. TK9. Calculations using irradiance and surface area. TK11. Power and energy production from solar arrays using irradiance. TK12. I-V curves, Efficiency, STC name plates of solar modules. TK13. Charge controller types i.e MPPT and non-MPPT technology. TK14. Battery types, battery storage capacity (A-h), Depth of Discharge (DoD), Cycle life. TK15. Inverter operations and types, Battery based invertors including hybrid type, Grid-tie invertors, module level power electronics (MLPE) micro-invertors. TK16. Over-current protection devices (fuses and circuit breaker) TK17. Understand voltage drops in electric circuits. System grounding and grounding conductors including surge protection. TK18. Mounting structures, cables, junction boxes and other components. TK19. Electrical and mechanical integration of PV components. TK20. Relevant tools for PV System design software. TK21. Relevant tools for shade analysis. TK22. Identify electrical symbols, interpret layout and wiring diagrams. TK23. Estimate material requirements, determine power demands and system sizing.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : RK1. Compliance aspects of the industry pertaining to Government regulations (licensing – that includes practicing licensing, occupational health and safety regulations, installation codes and standards for the solar industry).
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills The individual on the job must be able to: CS1. Read, understand and interpret internal/external documents such as manuals, health and safety instructions, memos, other company documents. CS2. Read from different sources- books, screens in machines and signages.

	CS3. Understand the various colour codes, as per standard electrical, mechanical and civil nomenclature. CS4. Work in a team (teamwork). CS5. Work under minimum or no supervision.
	Writing Skills
	The individual on the job must be able to : CS6. Fill up documentation applicable to one's role. CS7. Prepare reports. CS8. Prepare budgets.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to : CS9. Express statements or information clearly so that others (both internal and external) can hear and understand. CS10. Participate in and understand the main points of internal discussions. CS11. Respond appropriately to any queries.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Adhere to the organization's decision-making process. PS2. Comply with established procedures.
	Plan and Organise
	The individual on the job must be able to : PS3. Plan and organise work to meet deadlines. PS4. Work to achieve set goals and collaborate with others. PS5. Assemble appropriate team members for a particular project. PS6. Ensure that all necessary tools and materials available.
	Customer Centricity
	The individual on the job must be able to: PS7. Follow the set code of conduct and ethics of the organisation. PS8. Manage relationships with customers with intent on satisfying their requirements for service delivery.
	Problem Solving
	The individual on the job must be able to: PS9. Recognize problems and search for solutions. PS10. Choose best methods to complete assigned tasks. PS11. Consult relevant authorities when required.
	Analytical Thinking
	The individual on the job must be able to: PS12. Apply domain knowledge, observations and data to select course of action to perform tasks related to Solar PV Systems.
	Critical Thinking
	The individual on the job must be able to: PS13. Critically evaluate information/feedback obtained from customers, supervisor and co-workers to perform day to day activities. PS14. Handle unforeseen circumstances. PS15. Ask questions for better understanding.

UNIT 2 [This unit covers the skills, knowledge and understanding required by a solar PV system installer carry out civil, mechanical and electrical installation of a Solar PV System].

Unit No.	02
Unit Title	Installation of a Solar PV System
Description	This unit is about the installation of a Solar PV system.
Scope	This unit covers the following: • Equipment checklist • Component layout plan • Mounting of components • Conduiting and cabling • Grounding conductors and rods • Battery bank installation
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Prepare equipment checklist	To be competent, the individual must be able to: PC1. Prepare a list of all required equipment, manuals and tools. PC2. Prepare installation logistics. PC3. Inspect any physical damage to components before departure to the installation site.
Prepare component layout plan	To be competent, the individual must be able to: PC4. Prepare single line diagrams (SLD) for the PV system. PC5. Prepare equipment stencils for marking where to drill holes.
Mount components	To be competent, the individual must be able to: PC6. Decide module mounting type (roof, pole or ground). PC7. Mount components (inverter, charge controller, combiner boxes, distribution boxes, fuse boxes. etc.).
Install conduits and cabling	To be competent, the individual must be able to: PC8. Prepare conduit and select the critical path for cable. PC9. Install cable gutter/trunking/cable tray. PC10. Install conduit pipe and expansion fittings (where necessary). PC11. Mount over-current protection devices (OCPDs). PC12. Electrical integration of all mounted components. PC13. Use appropriate wire bending radius to avoid kinks.
Install grounding conductors and grounding rod	To be competent, the individual must be able to: PC14. Determine grounding conductor size. PC15. Electrically integrate all components that have metal chassis with the grounding conductor. PC16. Bond all grounding conductors to the grounding rod. PC17. Bond system grounding conductors and equipment grounding conductors to the same ground rod.

Install battery bank	To be competent, the individual must be able to: PC18. Confirm and install the rack/enclosure with spill containment (if required). PC19. Install pre-charged batteries and prepare battery terminals by cleaning them. PC20. Install all series and parallel interconnection cables as required and apply anti-oxidant material where necessary. PC21. Connect the batteries via appropriate OCPD (to the inverter and charge controller).
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company's Installation Policy. OK2. Company's Customer Support Policy. OK3. Company's documentation policy. OK4. Document information using appropriate corporate forms. OK5. Company's reporting structure. OK6. Organization culture. OK7. Workplace safety, identification, mitigation and prevention of safety hazards.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of : TK1. Appropriate tools for the job. TK2. Installation of solar array with the correct tilt angle and orientation for the particular location. TK3. The use of appropriate cable lugs for cable termination for battery termination. TK4. Appropriate ventilation and clearances of wall mounted components like invertors and charge controllers. TK5. Wire sizing for grounding and current carrying conductors. TK6. Grounding conductors and current carrying conductors must be in the same conduit. TK7. Workspace clearances from the solar equipment. TK8. Cable management. TK9. Electrical and mechanical integration of PV components.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : RK1. Compliance aspects of the industry pertaining to Government regulations (licensing – that includes practicing licensing, occupational health and safety regulations, installation codes and standards for the solar industry).
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills The individual on the job must be able to: CS1. Read, understand and interpret internal/external documents such as manuals, health and safety instructions, memos, other company documents. CS2. Read from different sources- books, screens in machines and signages.

	CS3. Understand the various colour codes, as per standard electrical, mechanical and civil nomenclature. CS4. Work in a team (teamwork). CS5. Work under minimum or no supervision.
	Writing Skills
	The individual on the job must be able to : CS6. Fill up documentation applicable to one's role. CS7. Prepare reports. CS8. Prepare budgets.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to : CS9. Express statements or information clearly so that others (both internal and external) can hear and understand. CS10. Participate in and understand the main points of internal discussions. CS11. Respond appropriately to any queries.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Adhere to the organisation's decision making process. PS2. Comply with established procedures.
	Plan and Organise
	The individual on the job must be able to : PS3. Plan and organise work to meet deadlines. PS4. Work to achieve set goals and collaborate with others. PS5. Assemble appropriate team members for a particular project. PS6. Ensure that all necessary tools and materials available.
	Customer Centricity
	The individual on the job must be able to: PS7. Follow the set code of conduct and ethics of the organisation. PS8. Manage relationships with customers with intent on satisfying their requirements for service delivery.
	Problem Solving
	The individual on the job must be able to: PS9. Recognize problems and search for solutions. PS10. Choose best methods to complete assigned tasks. PS11. Consult relevant authorities when required.
	Analytical Thinking
	The individual on the job must be able to: PS12. Apply domain knowledge, observations and data to select course of action to perform tasks related to Solar PV Systems.
	Critical Thinking
	The individual on the job must be able to: PS13. Critically evaluate information/feedback obtained from customers, supervisor and co-workers to perform day to day activities. PS14. Handle unforeseen circumstances. PS15. Ask questions for better understanding.

UNIT 3 [This unit covers the skills, knowledge and understanding required by a solar PV system installer carry out testing and commissioning of a Solar PV System].

Unit No.	03
Unit Title	Testing and Commissioning of Solar PV System
Description	This unit is about the testing and commissioning of an installed Solar PV system.
Scope	This unit/task covers the following: • System testing • Commission the system
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Testing the Solar PV system	To be competent, the individual must be able to: PC1. Perform visual inspection. PC2. Use appropriate PPE for the commissioning process. PC3. Identify all power sources and disconnect them. PC4. Test to ensure absence of voltage in order to prevent unintentional contact with energised parts. PC5. Test the voltage and polarity of all DC sources (PV arrays and battery banks). PC6. Test the AC input voltage of the grid or backup power sources (generator) if present. PC7. Verify system grounding and measure insulation resistance. PC8. Verify and calibrate data acquisition system.
Commission the Solar PV system	To be competent, the individual must be able to: PC9. Turn off and lock out/tag all disconnect boxes and breakers using appropriate equipment. PC10. Use appropriate PPE for the commissioning process. PC11. Remove all fuses except for those in the supply side connection. PC12. Turn off any additional breakers such as those in the combiner boxes. PC13. Torque check label, inspect and document as-built changes. PC14. Connect all power sources to the system. PC15. Begin the system start-up procedure as specified in the manual. PC16. Test system performance and function to confirm correct operation. PC17. Prepare testing and commissioning report.

Knowledge and Understanding (K)	
A. Organisation -al Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company's Installation Policy. OK2. Company's Customer Support Policy. OK3. Company's documentation policy. OK4. Document information using appropriate corporate forms. OK5. Company's reporting structure. OK6. Organization culture. OK7. Workplace safety, identification and mitigation of safety hazards.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of : TK1. Diagnostics/fault finding techniques. TK2. Correct application of signs and Labels. TK3. Understand log out/tag out procedure. TK4. Use of all required tools such as Infra red thermometer, multimeter, insulation resistance tester, torque wrench, clamp metre, pyranometer. TK5. Understanding of sun-path diagrams. TK6. Effect of blocking and bypass diodes. TK7. Basic functioning and Operation of different types of inverters and other electrical components. TK8. Installation work on a PV power system in accordance with relevant standards and regulations. TK9. Measurement of losses in a PV system at different points and interpretation of the results. TK10. Occupational health and safety (OHS) standards and associated risks when working on that particular site.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : RK1. Compliance aspects of the industry pertaining to Government regulations (licensing – that includes practicing licensing, occupational health and safety regulations, installation codes and standards for the solar industry).
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills
	The individual on the job must be able to: CS1. Read, understand and interpret internal/external documents such as manuals, health and safety instructions, memos, other company documents. CS2. Read from different sources- books, screens in machines and signage. CS3. Understand the various colour codes, as per standard electrical, mechanical and civil nomenclature. CS4. Work in a team (teamwork). CS5. Work under minimum or no supervision.
	Writing Skills
	The individual on the job must be able to : CS6. Fill up documentation applicable to one's role. CS7. Prepare reports.

	CS8. Prepare budgets.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to : CS9. Express statements or information clearly so that others(both internal and external) can hear and understand CS10. Participate in and understand the main points of internal discussions. CS11. Respond appropriately to any queries.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Adhere to the organization's decision making process. PS2. Comply with established procedures.
	Plan and Organise
	The individual on the job must be able to : PS3. Plan and organise work to meet deadlines. PS4. Work to achieve set goals and collaborate with others. PS5. Assemble appropriate team members for a particular project. PS6. Ensure that all necessary tools and materials available.
	Customer Centricity
	The individual on the job must be able to: PS7. Follow the set code of conduct and ethics of the organisation. PS8. Manage relationships with customers with intent on satisfying their requirements for service delivery.
	Problem Solving
	The individual on the job must be able to: PS9. Recognize problems and search for solutions. PS10. Choose best methods to complete assigned tasks. PS11. Consult relevant authorities when required.
	Analytical Thinking
	The individual on the job must be able to: PS12. Apply domain knowledge, observations and data to select course of action to perform tasks related to Solar PV Systems.
	Critical Thinking
	The individual on the job must be able to: PS13. Critically evaluate information/feedback obtained from customers, supervisor and co-workers to perform day to day activities. PS14. Handle unforeseen circumstances. PS15. Ask questions for better understanding.

UNIT 4 [This unit covers the skills and knowledge required by a solar PV installer to be proficient in operating and maintaining a Solar PV System].

Unit No.	04
Unit Title	Operations and Maintenance (O&M)
Description	This unit is about operating and maintaining a solar PV system
Scope	This unit/task covers the following: • Operating a solar PV system • Maintaining a solar PV system
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Operate a Solar PV system	To be competent, the individual must be able to: PC1. Check voltage, current and power at different level in the system. PC2. Monitor by way of measurements, power production from the PV array on a regular basis (hourly, daily, monthly and/or yearly). PC3. Record and store performance data from the SCADA system. PC4. Compute performance ratio and capacity utilization factor. PC5. Compare and analyse the actual power generation with the theoretical and simulation data and report any deviations PC6. Identify faults and analyse the cause.
Maintain Solar PV System	To be competent, the individual must be able to: PC7. Prepare preventive maintenance schedule. PC8. Inspect, clean and dust the components, check terminal firmness, measuring voltage and specific gravity of batteries (for flooded lead acid battery types). PC9. Check termination torques (re-torque terminals). PC10. Ensure proper functioning of grounding and lightening protection system. PC11. Document operation of all components by maintaining equipment log books. PC12. Analyse causes of faults in the solar PV system in a case of interruption or drop in power generation and ensure appropriate corrective action is taken. PC13. Analyse thermographic data to identify weak and loose connections or faulty solar modules. PC14. Ensure cleaning of work areas after completion of maintenance activities. PC15. Ensure structural integrity of civil foundations and module mounting structures. PC16. Ensure there is no shading on solar array during the day between 09:00-15:00 by periodically removing all obstacles including vegetation. PC17. Clean solar modules as per cleaning schedule.

	PC18. Ensure complete documentation of the job completion forms with signature of the supervisor as per relevant industry practice.
Knowledge and Understanding (K)	
A. Organisation-al Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: - OK1. Company's Installation Policy. - OK2. Company's Customer Support Policy. - OK3. Company's documentation policy. - OK4. Document information using appropriate corporate forms. - OK5. Obtain authorization from specified field safety officer and supervisor. - OK6. Company's reporting structure. - OK7. Organization culture. - OK8. Company's different department and concerned authority.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: - TK1. Operations of measuring instruments like solar radiation metre, thermographic imaging (infra-red camera), drones, I-V Tracer, Solar Pathfinder, and weather station. - TK2. SCADA system. - TK3. Maintenance philosophy- Run-to-failure (breakdown maintenance), Preventive (scheduled) maintenance, Predictive maintenance (PdM), Reliability-centered maintenance (RCM) and Condition Based Maintenance (CBM), Condition-based maintenance (CBM). - TK4. Performance ratio calculations.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : RK1. Compliance aspects of the industry pertaining to Government regulations (licensing – that includes practicing licensing, occupational health and safety regulations, installation codes and standards for the solar industry).
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills
	The individual on the job must be able to: - CS1. Read, understand and interpret internal/external documents such as manuals, health and safety instructions, memos, other company documents. - CS2. Read from different sources- books, screens in machines and signage. - CS3. Understand the various colour codes, as per standard electrical, mechanical and civil nomenclature. - CS4. Work in a team (teamwork). - CS5. Work under minimum or no supervision.
	Writing Skills
	The individual on the job must be able to :

	CS6. Fill up documentation applicable to one's role. CS7. Prepare reports. CS8. Prepare budgets.
	Oral Communication (Listening and Speaking skills) The individual on the job must be able to : CS9. Express statements or information clearly so that others (both internal and external) can hear and understand. CS10. Participate in and understand the main points of internal discussions. CS11. Respond appropriately to any queries.
B. Professional Skills	Decision Making The individual on the job must be able to: PS1. Follow organization rule-based decision making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise The individual on the job must be able to : PS3. Plan and organise work to meet deadlines. PS4. Work to achieve set goals and collaborate with others. PS5. Assemble appropriate team members for a particular project. PS6. Ensure that all necessary tools and materials available.
	Customer Centricity The individual on the job must be able to: PS7. Follow code of conduct. PS8. Manage relationships with customers with intent on satisfying its requirements for service delivery.
	Problem Solving The individual on the job must be able to: PS9. Recognize problems and search for solutions. PS10. Choose best methods to complete assigned tasks. PS11. Consult relevant authorities when required.
	Analytical Thinking The individual on the job must be able to: PS12. Apply domain knowledge, observations and data to select course of action to perform tasks related to Solar Photovoltaic Systems.
	Critical Thinking The individual on the job must be able to: PS13. Critically evaluate information/feedback obtained from customers, supervisor and co-workers to perform day to day activities. PS14. Handle unforeseen circumstances. PS15. Ask questions for better understanding.

UNIT 5 [This unit covers the skills and knowledge required by a solar PV installer to be aware of health and safety at the project site of a Solar PV System].

Unit No.	05
Unit Title	Safety, Health and Environmental Management
Description	This unit is about maintaining work safety at a solar PV system project site
Scope	This unit/task covers the following: • Establishment and following safe work procedure at workplace • Job hazard analysis • Hazard Identification • Hazard prevention and mitigation • Using and maintaining PPEs. • Safe and proper use of required tools and equipment. • Promotion of safety culture • Conforming to the requirements for environmental protection.
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Establish and follow safe work procedure	To be competent, the individual must be able to: PC1. Identify corporate policies required for workplace safety. PC2. Identify requirements for safe work area and create a safe work environment. PC3. Identify contact person when workplace safety policies are violated. PC4. Provide information about incidence/violation. PC5. Identify the location of First Aid materials and Fire assembly point.
Demonstrate understanding of Job hazard analysis	To be competent, the individual must be able to: PC6. Identify the tasks. PC7. Identify personal safety and/or work site hazards. PC8. Identify prevention and mitigation measures associated with any respective hazard.
Identify hazards	To be competent, the individual must be able to: PC9. Identify hazards common in PV project such as working at heights, falling in trenches, electrical shock, overhead impacts, heat stress and UV radiation exposure, chemical hazards from batteries, sharp objects, dust, noise, slippery surfaces, electrical arc flash.
Prevent and mitigate hazards	To be competent, the individual must be able to: PC10. Use the lock out/tag out procedure during commissioning and maintenance. PC11. Be aware of the potential overhead impacts and must wear hard-hat. PC12. Wear appropriate PPEs (gloves and safety boots) to avoid injury. PC13. Use full body harness when working at heights.

	PC14. Use appropriate PPEs (safety glasses, boots, gloves and acid proof work suits) to avoid corrosive substances from batteries. PC15. Use appropriate PPEs (earplugs) to avoid deafening noise. PC16. Use appropriate PPEs (sunscreen), stay hydrated, erecting a shade structure, taking breaks in a shade to avoid heat stress and UV exposure. PC17. Use appropriate PPEs (dust masks) and spray water on dusty grounds. PC18. Re-assess risk control measures, in accordance with changed work practices and/or site conditions and undertake alterations. PC19. Inspect/install fall protection and perimeter protection equipment ensuring adequacy for work and conformance to regulatory requirements. PC20. Identify approved methods of moving tools and equipment to work area and minimize potential risks associated with the hazards.
Demonstrate safe and proper use of required tools and equipment.	To be competent, the individual must be able to: PC21. Select tools, equipment and testing devices needed to carry out the work. PC22. Demonstrate safe and proper use of required tools and equipment.
Use and maintain PPEs.	To be competent, the individual must be able to: PC23. Identify the PPEs required for specific locations on-site and activity. PC24. Identify expiry dates of specified equipment. PC25. Demonstrate safe and accepted practices for personal protection.
Promote safety culture	To promote safety culture, the individual must be able to: PC26. Only allow authorised, competent and qualified personnel to work at the site. PC27. Report accident cases. PC28. Protect whistle blowers. PC29. Think before acting. PC30. Never work alone however, too many people in one place can be a hazard. PC31. Rewards or sanctions for promotion or violation of safety practices respectively.
Conforming to the requirements for environmental protection	To be competent, the individual must be able to: PC32. Read, interpret and implement the environmental policies of the organisation. PC33. Read, interpret and implement the environmental regulations of the organisation regarding PV installations. PC34. Identify environmental hazards and assess risks of all aspects of operations for plant and equipment.

Knowledge and Understanding (K)	
A. Organisation -al Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: - OK1. Company's Installation Policy. - OK2. Company's Customer Support Policy. - OK3. Company's documentation policy. - OK4. Document information using appropriate corporate forms. - OK5. Company's authorization processes between field .safety officers and supervisors. - OK6. Company's reporting structure. - OK7. Organization culture. - OK8. Company's different department and concerned authority - OK9. Company's Health and Safety Policies and Regulations. - OK10. Company environmental policies and regulations.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: - TK1. Relevant PPE required at a solar PV project site. - TK2. Relevant standards and regulations for installation of solar PV system installation in Zambia. - TK3. Occupational health and safety (OHS) standards for installation of Solar PV. - TK4. Identifying existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous or dangerous to employees. - TK5. Environmental Risk/Impact Assessment – identify risks that components of a PV system can cause to the environment and human health. - TK6. End-of-Life Environmental Management of PV value chain. - TK7. PV panel composition - TK8. Classification of PV system waste - to identify what to recycle and what to dispose of.
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of : RK1. Compliance aspects of the industry pertaining to Government regulations (licensing – that includes practicing licensing, occupational health and safety regulations, installation codes and standards for the solar industry).
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills
	The individual on the job must be able to: - CS1. Read, understand and interpret internal/external documents such as manuals, health and safety instructions, memos, other company documents. - CS2. Read from different sources- books, screens in machines and signage. - CS3. Understand the various colour codes, as per standard electrical, mechanical and civil nomenclature. - CS4. Work in a team (teamwork). - CS5. Work under minimum or no supervision.
	Writing Skills

B. Professional Skills	The individual on the job must be able to : CS6. Fill up documentation applicable to one's role. CS7. Prepare reports. CS8. Prepare budgets.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to : CS9. Express statements or information clearly so that others (both internal and external) can hear and understand. CS10. Participate in and understand the main points of internal discussions. CS11. Respond appropriately to any queries.
	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to : PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery. .
	Problem Solving
	The individual on the job must be able to: PS7. Recognize problems and search for solutions. PS8. Choose best methods to complete assigned tasks. PS9. Consult relevant authorities when required.
	Analytical Thinking
	The individual on the job must be able to: PS10. Apply domain knowledge, observations and data to select course of action to perform tasks related to Solar Photovoltaic Systems.
	Critical Thinking
	The individual on the job must be able to: PS11. Critically evaluate information/feedback obtained from customers, supervisor and co-workers to perform day to day activities. PS12. Handle unforeseen circumstances. PS13. Ask questions for better understanding.

5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS

Materials and tools needed to perform this job include, but not limited to; Computer; Personal protective equipment (PPE) (work suit, safety boots, hard hat, safety goggles, acid proof coat, ear plugs, safety belt, safety gloves etc); Electrical multimeter; Hammer; Insulation resistance tester; Plum bob; Pressure gauges; Spirit level; Threading equipment; Welding machine; Grinder; Drilling machine; Wheelbarrow; First aid kit; Locks and tag out kits; Safety signages; Safety tools such as fire extinguisher, barricades, company's health and safety policy/procedure; Zambian Standards and recognised codes of practices applicable to PV installers; OHS safety rules; Weather station; Hydrometer; Data logger; Infra-red camera; I-V tracers, PV design software; Solar pathfinder; Adjustable spanner; Shovel; Cable tester and locator; Clamp meter; Combination spanner; Ground resistance tester; Insulation tester; Loop tester; Pipe wrench; Pliers; Thread tape; Weld testing tools; Tools box; csLadder

6. DILEMMAS/CHALLENGES AND COMPLEXITIES FOR A JOB HOLDER

Dilemmas associated with the job of a Solar PV Installer include: Working in remote areas; Risk of snakes and mosquito bites; Risk of slipping and falling in trenches; Risk of electrocution; Handling and disposing corrosive material; Exposure to extreme weather such as hot and cold conditions; Understanding different customer interests; Working at height; Working in confined spaces; Lifting heavy equipment and tools; Working away from home; long working hours; Pressure from supervisors and colleagues; Time pressure to complete tasks; Pressure from government regulators; Working in noisy, wet and dusty environments, etc.

6.1 Alternative Choices (Solutions) to Dilemmas and Complexities

Solutions to dilemmas include: Preparing adequately; Plan for preventions and emergency services; Provide PPE and good housekeeping; Observe electrical safety procedures; Use PPE and follow environmental protection procedures; Prepare for various weather conditions; Apply customer relations; Provide ladders, working platform and scaffolding; Use appropriate lifting equipment Follow procedures for working in confined space; Provision for leave and rest days; exercising regularly to maintain physical fitness, participating in workplace safety sensitisation and awareness meetings/training sessions, planning your daily tasks, reading and researching new ways to handle work stress, adhering to company's safety and standard operating procedures at all times, consulting extensively within and outside one's department/team on safety and other issues.

7. WORKING CONDITIONS/ENVIRONMENT

Working conditions include: working in cold; Hot and wet condition; Working at heights; Stand/walk for long hours, Working in laboratory environment, Office buildings, Working in shifts, Areas that are noisy and dusty, Areas with limited lighting and ventilation. Working in confined spaces, working in explosives environment; Wild and remote areas.

8. PARTIES INVOLVED/INTERACTING WITH THE JOB HOLDER OR TRAINEE

8.1 Internal/Within the Organization

Parties involved/interacting with the job holder who are internal to the organization include: Supervisors and subordinates; Stores and Procurement; personnel and other employees (e.g. Finance and HR personnel, Architects, Surveyors, Engineers: Civil, Building, Electrical and Mechanical), Trainers, Safety team, etc.

8.2 External/Outside the Organisation

Parties involved/interacting with the job holder who are external to the organization include: Consultants and Contractors(customers/clients); Trainers: Government regulators; Suppliers of equipment/tools/consumables; Fellow Solar PV Installers from other companies; Labour unions; OHS associations; Local community etc.

9. PHYSICAL DEMANDS ON THE BODY

- Physique to sustain strenuous conditions such as climbing heights and lifting heavy things;
- Be able to walk and stand for long periods of time;
- Bend, stretch, twist, or reach out;
- Be able to lift relatively heavy materials, tools and equipment;
- Be able to use fingers, hands and feet with ease to complete the assigned task (dexterity).

ANNEX A

Criteria for Assessments based on this NOS

A.1 Guidelines for Assessment

- A.1.1** Criteria for assessment for curricula and learning programmes based on this NOS will be created by curricula and programmes developers. Each Performance Criteria (PC) will be assigned marks proportional to its importance in the NOS. Curricula and programmes developers will also lay down proportion of marks for theory and practical skills for each performance criteria, giving more weight to practical skills.

There shall be allocated the 'Total Mark', which will be the sum of all marks in each Unit, distributed across the number of PCs in that particular Unit. The 'out of' mark will be the mark allocated to each PC, which will be shared between theory and skills practical assessments.

- A.1.2** Awarding/assessment bodies or institutions and other users of the NOS will create unique question papers for the theory part and evaluations for skill practical part for their respective candidates.

ANNEX B NOS Version Control

This Annex gives details necessary for the tracking of the NOS versions based on the number of revisions.

NOS Code	NOS.SPVI.01		
ZQF Level	Level 6	Version Number	01
Sector	Energy	Date of Approval	September 2021
Sub Sector	Renewable Energy	Date of Last Review	N/A
Occupation	Solar PV Installer	Date of Next Review	September 2026

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